

MULTICORODO PROJECT Advanced ESP32-S3 Handheld Multi-Tool Device

Abstract

MULTICORODO is an advanced handheld embedded system developed around the ESP32-S3 microcontroller. The project combines wireless communication, AI interaction, NFC/RFID, infrared control, multimedia playback, secure wallet functionality, GPS tracking, and IoT control into a single compact portable device. The objective of the project is to build a modular cyber-tool platform capable of replacing multiple independent gadgets with one integrated embedded system.

1. Purpose of the Project

The main purpose of MULTICORODO is to create a portable embedded platform that integrates communication, security, automation, and control systems into a unified device. The project demonstrates advanced embedded engineering concepts including wireless communication, modular architecture, battery-powered design, secure firmware handling, and real-time user interfaces. The system is designed for experimentation, IoT control, smart automation, multimedia handling, and portable cyber-tool applications.

2. Core Controller – ESP32-S3

Component Summary Table

Component	Purpose
ESP32-S3	Main microcontroller and processing unit
ILI9341 TFT	Graphical user interface display
PN532	NFC and RFID communication
CC1101	Sub-GHz RF communication
SIMA7670E	4G LTE, GPS, SMS, calls
MAX98357A	Audio amplification
TP4056	Battery charging module
LiPo Battery	Portable power supply

The ESP32-S3 acts as the central processing unit of the project. It manages all modules including display rendering, communication handling, networking, storage, and device logic.

Main Features:

- Dual-core 240 MHz processor
- Integrated WiFi and Bluetooth
- 16 MB Flash and 8 MB PSRAM support
- High-speed SPI, UART, I2C communication
- Suitable for multitasking and graphical interfaces

3. Display and User Interface

The device uses a 2.8-inch ILI9341 SPI TFT display with 320x240 resolution. The graphical interface provides menus, status information, NFC data, AI responses, media controls, and phone functions.

User Input Devices:

- KY-040 Rotary Encoder for navigation
- 4x4 Matrix Keypad for command input and PIN entry

4. Communication Systems

MULTICORODO integrates several communication technologies into one platform.

SIMA7670E 4G LTE Module:

- Internet connectivity
- Voice calling
- SMS messaging
- GPS functionality

CC1101 Sub-GHz RF Module:

- RF signal capture
- Replay transmission
- Frequency analysis
- Wireless experimentation

5. NFC and RFID System

The PN532 NFC/RFID module operates at 13.56 MHz and supports communication with NFC tags and RFID cards.

Functions Provided:

- NFC tag reading
- NFC tag writing
- Card identification
- Automation triggers using NFC interactions

6. Infrared Remote System

The infrared subsystem consists of an IR receiver and IR transmitter LED. It enables universal remote control functionality.

Capabilities:

- Learn remote commands
- Store IR signals
- Replay commands
- Control TVs, air conditioners, and other IR devices

7. Audio and Multimedia System

The MAX98357A I2S amplifier is connected to a speaker for audio output.

Features:

- MP3 playback
- Voice prompts
- AI voice responses
- Call audio handling
- Notification sounds

8. Storage and File Management

MULTICORODO supports external storage through USB or microSD-based systems.

Storage Functions:

- Music storage
- RF and NFC data storage
- Logs and configuration files
- Wallet backups

9. Security Architecture

Security is one of the major design objectives of MULTICORODO.

Security Features:

- Device PIN protection
- Encrypted wallet system
- Air-gapped wallet mode
- OTA password verification
- TOTP-based authentication

10. Power System

The device is powered using a rechargeable LiPo battery with charging and voltage regulation circuitry.

Power Components:

- 3.7V LiPo battery
- TP4056 charging module
- Voltage regulation system
- Capacitor stabilization for 4G current spikes

11. Wiring Architecture

MULTICORODO uses a modular wiring architecture.

Main Communication Interfaces:

- SPI: TFT display, CC1101, PN532
- UART: SIMA7670E 4G module
- I2S: MAX98357A audio amplifier
- GPIO: IR transmitter, keypad, rotary encoder

The modular structure allows easier debugging, upgrades, and maintenance.

12. Major Features of MULTICORODO

Main System Features:

- AI assistant interaction
- Phone calling and SMS
- GPS location tracking
- RF signal analysis
- NFC/RFID toolkit
- Universal IR remote
- Music player
- Secure crypto wallet mode
- Smart Hall IoT integration
- Portable handheld graphical interface

13. Conclusion

MULTICORODO represents a highly advanced embedded systems project integrating multiple technologies into one compact device. The project demonstrates strong engineering concepts including modular architecture, secure system design, communication systems integration, embedded user interfaces, and portable power management.

The platform has future potential for expansion into professional cybersecurity tools, IoT management systems, smart

automation controllers, and secure portable communication platforms. The project also demonstrates the capability of modern microcontrollers such as the ESP32-S3 to support complex multifunction embedded systems.